

High-Load-Triggered Nanoscale Superlubricity in Long-Chain Borate Ionic Liquids

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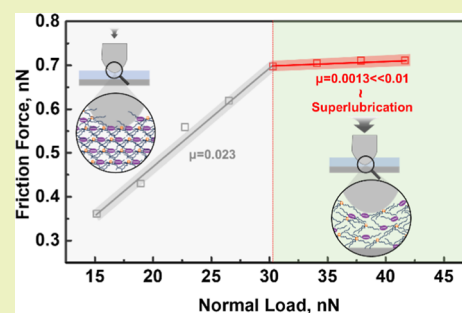
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Supporting Information

ABSTRACT: Achieving nanoscale superlubricity with ionic liquids (ILs) in the absence of external fields or additives remains a major challenge in tribology. Here, we investigate the nanoscale friction of a homologous series of three borate-based ILs (denoted A4, A8, and A12) with a varying anion alkyl-chain length $[A_4BMB]^-$, $[A_8BMB]^-$, $[A_{12}BMB]^-$, and a common cation trioctyldecylammonium ($[N_{88812}]^+$) on graphite surfaces, using AFM, in situ AFM-IR, angle-resolved XPS (AR-XPS), and nonequilibrium molecular dynamics (NEMD) simulations. Both A4 and A8 show stable superlow friction across the load range ($\mu \approx 0.0032$ and 0.0068 , respectively). A12, however, demonstrates a frictional transition triggered by an increasing load. The friction $\mu \approx 0.023$, obtained at a low load, drops drastically to $\mu \approx 0.0013$ once the normal load exceeds ~ 30 nN (≈ 2.4 GPa) and enters a clear superlubric state. As neither AFM-IR nor AR-XPS reveals any tribochemical transformation during this transition, it appears to be a purely physical, load-induced structural reorganization of the interfacial ion layers. Further studies show near-homogeneous cation/anion distributions in A12 at the interface, while NEMD simulations can identify load-dependent reorientation of the long anion alkyl chains that reduces interfacial locking and shear resistance. Our results show a purely mechanical unlocking pathway to nanoscale superlubricity in ILs and suggest that long-chain borate anions can be used for extreme-pressure lubrication in micro/nano electromechanical systems.

KEYWORDS: friction transition, superlubricity, high load, reorientation, AFM-IR



INTRODUCTION

Superlubricity, defined as a friction regime with a friction coefficient (μ) below 0.01, represents a paradigm shift in tribology, with immense potential for reducing global energy losses and wear in mechanical systems.¹ It has been widely observed in two-dimensional layered solids,^{2,3} such as graphite and MoS_2 , where incommensurate lattice structures allow sliding with vanishing resistance. However, these systems often require harsh conditions, including specific atmosphere, atomically flat surfaces, etc.,^{4–7} limiting their practical applications.

Compared with solid-state lubricants, liquid lubricants, especially ionic liquids (ILs), offer greater versatility under ambient conditions. ILs are defined as salts composed of organic cations and organic or inorganic anions with melting points below 100 °C and were first proposed as potential lubricants in 2001.⁸ ILs exhibit excellent thermal and electrochemical stability, strong surface affinity, and tunable molecular structures, making them attractive for boundary lubrication.^{9–11} Some ILs and their analogues have demonstrated robust macroscale superlubricity through tribochemical/physical adsorption ion layers,^{1,12,13} hydrodynamic lubrication,¹⁴ etc. Despite the progress in IL-based superlubrication at the macroscale, the ongoing miniaturization of

micro- and nanoelectromechanical systems (MEMS/NEMS) has led to a dramatic increase in the surface-to-volume ratio, rendering friction losses a critical bottleneck for maintaining performance and reliability at the nanoscale.¹⁵

The nanoscale realm hosts a broad spectrum of intermolecular forces that compete to make new behavior of ILs emerge via the interaction of the IL with the contacting surfaces.^{16,17} Most reported IL-based superlubric systems are primarily enabled by electric-field-induced reorganization of the interfacial ion layers.^{18–21} Recently, He et al. achieved nanoscale superlubricity by regulating interfacial ion structuring of an imidazolium-based IL on a graphite surface through the addition of low-surface-energy perfluorocarbons, which promotes the monolayered IL formation at the surface.²² This raises an important question: in the absence of external electric fields or molecular additives, can superlubricity still be realized for ILs at the nanoscale?

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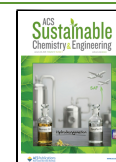


Table 1. Full Name, Abbreviation, and Cation/Anion Structures of the Three ILs with the Same $[N_{88812}]^+$ Cation but Differently Chained Anions, Utilized in This Work

Full name	Trioctyldodecylammonium bis(4-butylmandelato)borate	Trioctyldodecylammonium bis(4-octylmandelato)borate	Trioctyldodecylammonium bis(4-dodecylmandelato)borate
Abbreviation	$[N_{88812}][A_4BMB]$, A4	$[N_{88812}][A_8BMB]$, A8	$[N_{88812}][A_{12}BMB]$, A12
Molecular structure	The figure displays the molecular structures of three ionic liquids (ILs) with the common cation $[N_{88812}]^+$ and different anions. The cation is a long-chain ammonium ion with three octyl groups and one dodecyl group, with a total length of 20.7 Å and a width of 14.5 Å. The anions are bis(4-n-alkylmandelato)borate ions, where n = 4, 8, and 12. The dimensions for the anions are as follows: $[A_4BMB]^-$: 6.6 Å (width), 8.2 Å (height), 18.5 Å (length) $[A_8BMB]^-$: 6.3 Å (width), 9.8 Å (height), 25.4 Å (length) $[A_{12}BMB]^-$: 7.0 Å (width), 12.4 Å (height), 26.7 Å (length)		

One promising route lies in leveraging mechanically induced confinement to alter the interfacial ion arrangement,^{11,16,23} i.e., applying sufficiently high normal loads between two sliding surfaces to compress the IL into a highly confined interfacial environment.^{24,25} Such load-dependent confinement has been shown to reorganize interfacial ion layers, probably unlocking superlubricity in the absence of external stimuli. For example, a previous study has shown that increasing normal load can induce interfacial ion reorganization and reduce friction in confined IL films, even resulting in a counterintuitive negative friction–load dependence.²⁶

Therefore, in this work, we aim to explore whether high-load-induced nanoscale confinement can be harnessed to trigger superlubricity in IL systems without external stimuli. To this end, a series of borate-based ILs sharing the same ammonium cation ($[N_{88812}]^+$) but differing in bis(4-*n*-alkylmandelato)borate anion alkyl chain length, namely, $[A_4BMB]^-$, $[A_8BMB]^-$, and $[A_{12}BMB]^-$, were systematically investigated. We found that only the IL with the longest anion— $[A_{12}BMB]^-$, bearing two dodecyl (C12) side chains—exhibits a distinct frictional transition: while it shows relatively high friction at low normal loads, it undergoes a sharp drop into the superlubricity regime under higher loads (~ 2.4 GPa). In contrast, the two shorter-chain ILs, $[N_{88812}][A_4BMB]$ and $[N_{88812}][A_8BMB]$, maintain relatively stable frictional behavior across the entire load range without a clear transition behavior. Spectroscopic analyses confirm that no interfacial chemical reactions occur during the superlubricity transition of the longest-chain IL, $[N_{88812}][A_{12}BMB]$. Molecular dynamics simulations suggest that the observed superlubricity is instead driven by load-induced structural reorganization of the interfacial ion layers. These findings uncover a previously underexplored mechanical unlocking pathway to superlubricity transition, driven by ion rearrangements, and offer new insights for the design of sustainable, adaptive lubrication in extreme nanoscale environments.

EXPERIMENTAL SECTION

Materials

Diethyl malonate, 1-bromobutane, 1-bromooctane, 1-bromododecane, tri-*n*-octylamine, lithium hydroxide, and boric acid were analytical grade and received from Shanghai Macklin Biochemical Technology Co., Ltd., for synthesizing the three ILs with varying ion sizes (cf. Table 1): (1) trioctyldodecylammonium bis(4-butylmandelato)borate ($[N_{88812}][A_4BMB]$, A4); (2) trioctyldodecylammonium bis(4-octylmandelato)borate ($[N_{88812}][A_8BMB]$, A8); (3) trioctyldodecylammonium bis(4-dodecylmandelato)borate ($[N_{88812}][A_{12}BMB]$, A12). Detailed synthesis procedure and characterization by 1H , ^{11}B NMR spectra of A4, A8, and A12 (Figure S1) can be found in the Supporting Information. The SNL probes (silicon nitride, radius of 2 nm) for friction measurements were from Bruker. The highly oriented pyrolytic graphite (HOPG) was purchased from Mikromasch as the supporting substrate for the ILs. Note that graphite is widely used as a standard model tribological surface because it is atomically smooth and chemically inert, enabling reproducible AFM friction measurements and direct comparison with extensive literature. Additionally, graphite and graphene-based materials are increasingly integrated into MEMS/NEMS devices, like sensors and actuators,²⁷ making them highly relevant substrates for nanoscale lubrication studies.

Preparation of IL Film Supported by Graphite

To prepare IL films on the surface of graphite, the ILs were dissolved in acetone at a concentration of 10^{-3} mL of ILs/mL of acetone. A droplet of the as-prepared IL solution (1 μ L) was deposited onto the graphite substrate (area = 1 cm²) and then allowing the acetone to completely evaporate under vacuum at 35 °C for 12 h.

Force Measurements

The friction force measurements were performed by a Bruker Dimension Icon atomic force microscopy (AFM) in contact mode at ambient conditions to examine the ion dynamics of the ILs confined to graphite surfaces, with SNL probes in a scan range of 2 μ m $\times$ 2 μ m (scan rate = 2 Hz), under normal loads from 0 to 3.5 V. The friction force calibration was performed based on Liu's method.²⁸

AFM-IR

AFM-IR measurements were conducted using a Bruker Dimension IconIR system, which integrates atomic force microscopy with infrared spectroscopy (AFM-IR) for spatially resolved chemical characterization. The IL films supported on graphite were probed under varying normal loads by using a contact-mode AFM probe

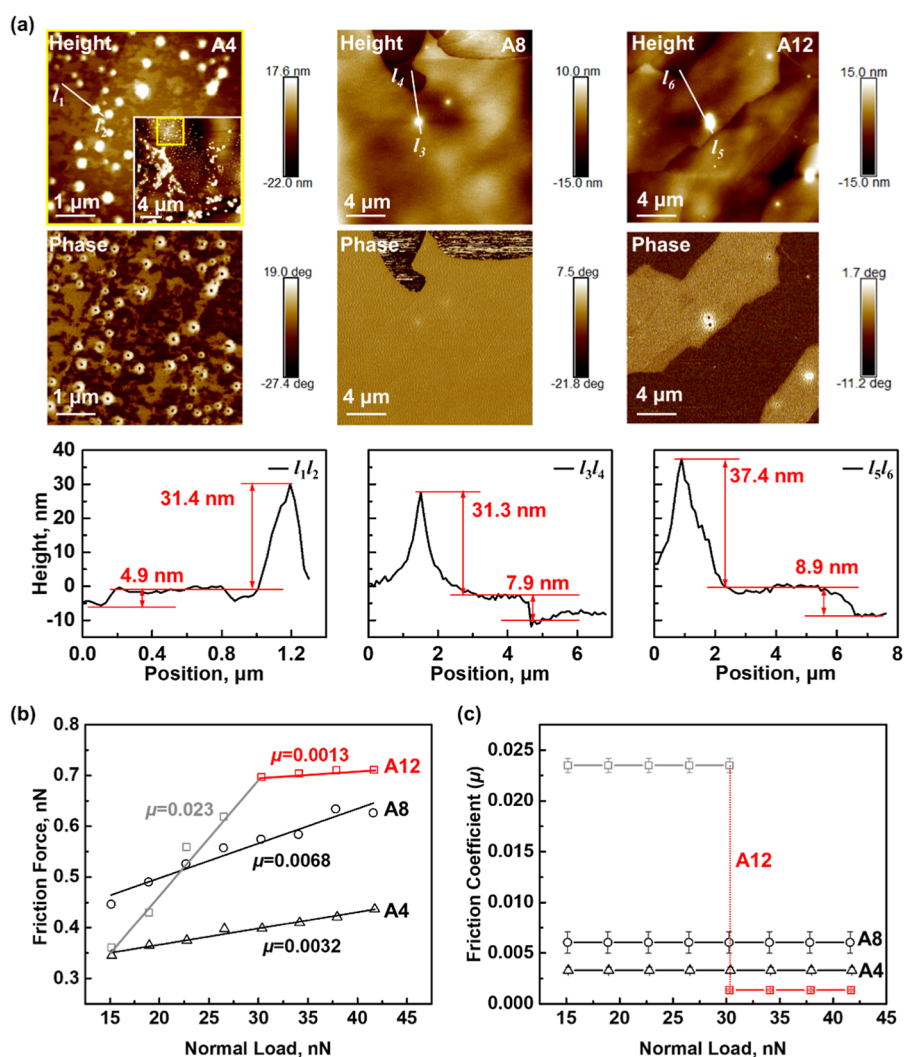


Figure 1. (a) Surface morphologies (upper), phase contrasts (middle), and cross-sectional profiles (bottom) of A4, A8, and A12 ILs at graphite surfaces. The A4 topography image is a magnified view of the yellow-boxed area in the inset. (b) Typical friction force vs normal load relationship for the silicon nitride AFM probe sliding at IL–graphite interfaces. The friction coefficient μ is determined by the slope of the fitting line, following Amontons’ law of friction. (c) The corresponding averaged regressed friction coefficient as a function of the varying normal loads.

equipped with photothermal IR detection. The applied normal load was precisely controlled via the AFM module (from -2 V to 3 V) to simulate frictional contact conditions. IR spectra were collected at selected points on the IL surface to monitor the load-induced chemical changes at the sliding interface.

X-ray Photoelectron Energy Spectrum (XPS) Measurements

XPS depth profiles were acquired to analyze the element distribution of the ILs at varying distances from the underlying graphite surfaces. XPS measurements were performed using a Thermo Scientific K-Alpha X-ray photoelectron spectrometer with monochromatic Al K α X-ray.

Simulation

In the nonequilibrium molecular dynamics (NEMD) calculations, the AFM probe was modeled as a capped single-walled carbon nanotube (SWCNT) with the cap radius of 0.5 nm. The CNT probe is a common MD proxy for sampling interfacial forces.²⁹ Although our experiments use a silicon nitride probe, this CNT representation is standard in AFM-MD due to tractability and the ability to use small, well-defined apex radii,^{29,30} and it has been used alongside silicon/silicon nitride experimental probes in combined AFM–MD studies.³¹ The simulation box had the dimension of $4.26 \times 4.43 \times 17.0$ nm³ and contained 106 [N₈₈₈₁₂][A₁₂BMB] (A12) molecules. The bottom of

the simulation box was modeled via a fixed single-layer graphene surface. The NEMD friction simulation protocol started by placing the capped SWCNT probe 2.0 nm into the IL/graphene system to ensure that the probe was surrounded by IL molecules (IL thickness of ~ 11.0 nm). Varying external normal loads were then applied to bring the probe into intimate contact with the IL/graphene interface, and the probe slid laterally on the IL/graphene interface at a constant velocity of 1.0 m/s.

RESULTS AND DISCUSSION

Nanoscale Frictional Behavior Tuned by Anion Chain Length and Applied Load

As shown in Figure 1(a), the surface morphologies and phase contrasts of the three borate ILs sharing the same cation but different anions, i.e., [N₈₈₈₁₂][A₄BMB], [N₈₈₈₁₂][A₈BMB], and [N₈₈₈₁₂][A₁₂BMB] (denoted as A4, A8, and A12, respectively), on graphite exhibit distinct dependence on the anion alkyl chain length. The short-chain A₄ film displays discontinuous, island-like domains with large height variations (~ 4.9 – 31.4 nm), indicative of poor surface wetting and weak interfacial interactions. In contrast, the medium-chain A₈ system forms the most uniform and extended coverage (~ 7.9 – 31.3 nm),

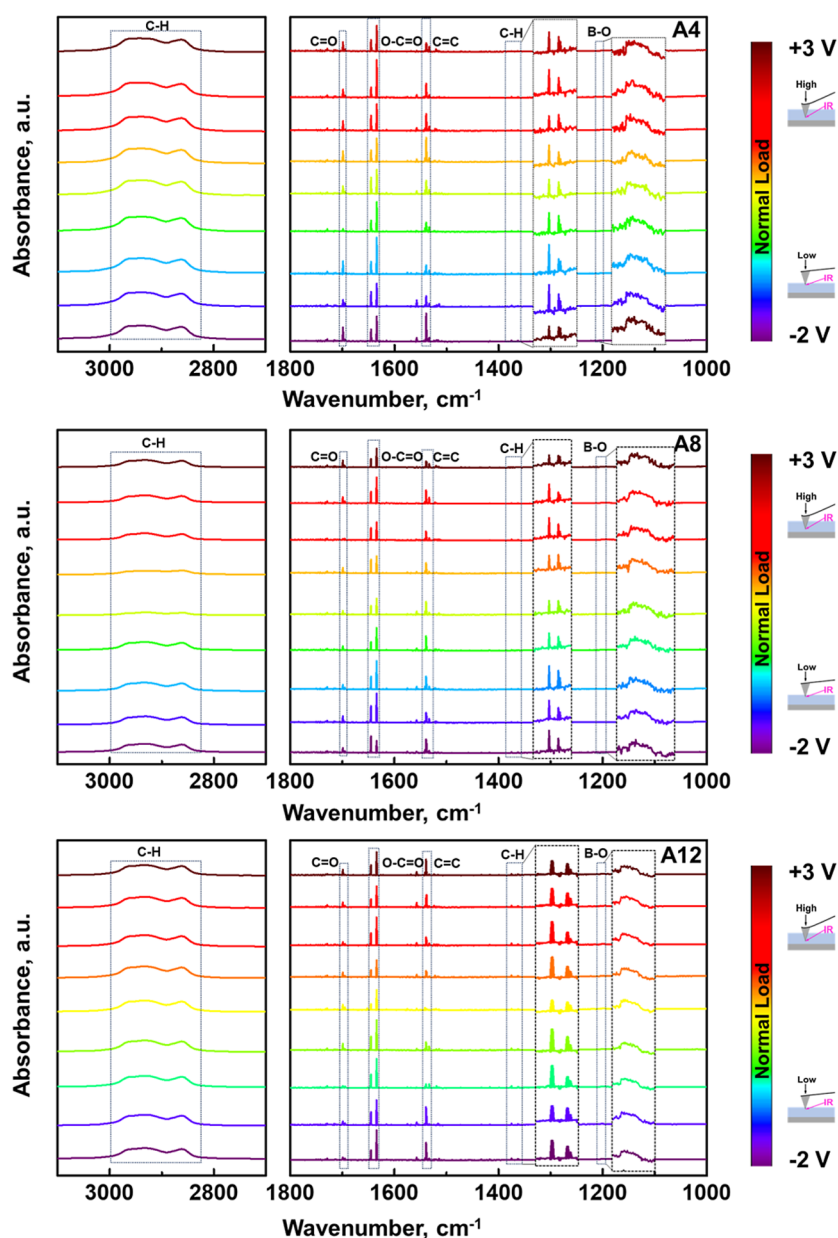


Figure 2. AFM-IR spectra (left: 3000–2800 cm^{-1} , right: 1800–1000 cm^{-1}) of A4, A8, and A12 recorded under varying normal loads (–2–3 V, where –2 V corresponds to a lower load, and 3 V corresponds to a higher load). The color gradient indicates increasing normal load.

suggesting enhanced surface wetting and better molecular packing. Interestingly, although the A₁₂ system possesses the longest alkyl tails and exhibits the largest thickness (~8.9–37.4 nm), its morphology reveals partially aggregated domains rather than fully homogeneous layers. This nonmonotonic wetting behavior likely arises from competing effects of increased van der Waals attraction, which promotes ion adsorption, and excessive chain length, which leads to self-association and reduced spreading. The underlying interactions are probably mainly attributed to a combination of CH– π interactions and van der Waals forces between long alkyl chains and graphite. The sp²-hybridized graphite surface, being rich in π -electron density, exhibits weak negative character and tends to attract the initial adsorption of the long alkyl chains of the IL cations via CH– π affinity.³² In addition, previous experimental and theoretical studies have demonstrated that ILs with long alkyl chains tend to form ordered nanostructures

at graphite³³ and gold^{34,35} interfaces via such van der Waals interactions. In our system, both the bulky trioctyldodecylammonium cation, [N₈₈₈₁₂]⁺, with multiple alkyl tails and long-tailed anions provide sufficient van der Waals interactions with the graphite basal plane, promoting IL adsorption and surface wetting. The initial interface formation is likely dominated by the adsorption of the cation layer, which is followed by reorientation and packing of both cations and anions to minimize interfacial energy.

To investigate the frictional response of borate-based ILs, we conducted friction force measurements using an AFM probe sliding on IL films supported on graphite substrates. Friction force versus normal load curves for the three ILs are shown in Figure 1(b), and the corresponding friction coefficients are summarized in Figure 1(c). Under low-to-moderate loads (15–30 nN), all ILs displayed linear frictional behavior following Amontons' law. Both short-chain A4 and medium-

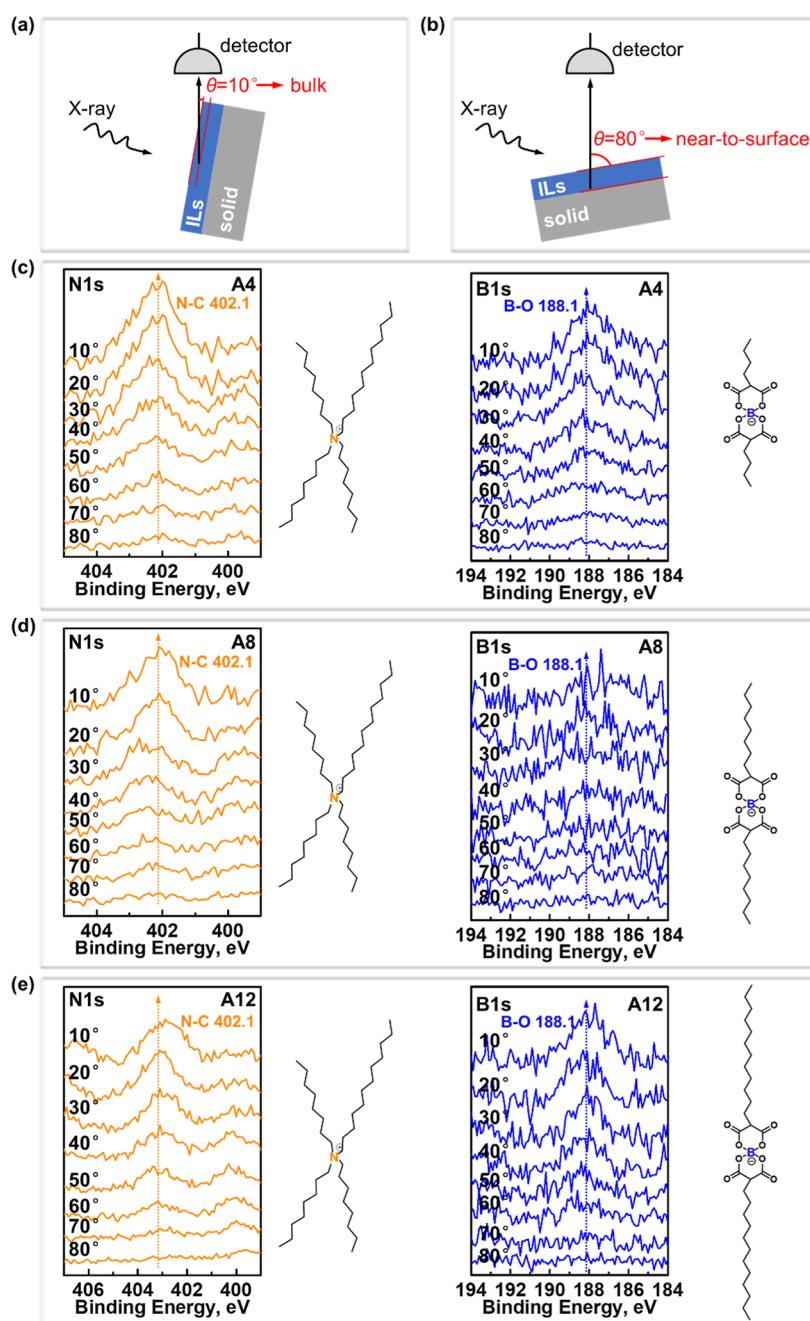


Figure 3. Angle-resolved X-ray photoelectron spectroscopy (AR-XPS) analysis of the chemical state of IL species. Schematic illustration of AR-XPS detection geometry: (a) small emission angles (e.g., 10°) probe deeper bulk-like ions, while (b) large angles (e.g., 80°) are surface-sensitive and detect near-to-surface ionic layers. High-resolution N 1s and B 1s spectra of (c) A4, (d) A8, and (e) A12, respectively, at emission angles of 10° – 80° .

chain A8 ILs exhibited consistently low friction coefficients of $\mu = 0.0032$ and 0.0068 , respectively, over the entire load range. These values fall within the superlubricity regime ($\mu < 0.01$), indicating their ability to maintain extremely low shear resistance at the nanoscale. Although the A8 system exhibits the best initial wetting behavior on graphite, as evidenced by its uniform morphology and phase homogeneity in Figure 1(a), its friction coefficient is higher than that of less-wetting A4. This may be attributed to the discontinuous coverage of the A4 film, which reduces real contact area and minimizes interfacial locking, facilitating the reduction of friction coefficient.^{36–38}

Instead, the A12 system with the longest anion, bearing two dodecyl side chains, despite its less uniform initial coverage, displays the most dramatic reduction in friction coefficient under high load (Figure 1b,c), ultimately entering the superlubricity regime ($\mu \approx 0.0013$). This apparent discrepancy suggests that the initial surface wetting alone is not the sole determinant of frictional performance. Specifically, A12 displays a distinctive load-responsive frictional transition: at lower loads ($\leq 30 \text{ nN}$), it exhibits a significantly higher friction coefficient ($\mu \approx 0.023$), which is outside the superlubricity threshold. However, upon increasing the load beyond $\sim 30 \text{ nN}$ ($\approx 2.4 \text{ GPa}$ for a 2 nm -radius probe), a sharp drop in μ to

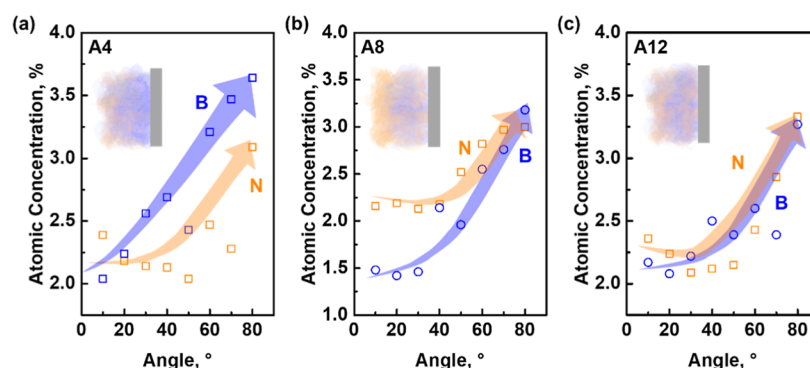


Figure 4. Depth-dependent atomic concentrations of N (cation marker) and B (anion marker) derived from AR-XPS at various emission angles (10° to 80°) for (a) A4, (b) A8, and (c) A12. Increasing atomic percentages with larger angles indicate surface enrichment. Insets show the cation (orange) and anion (blue) distribution at the interface.

0.0013 was observed, well within the superlubricity regime. This behavior indicates a load-triggered transition from high friction to superlubricity, a phenomenon that is absent in A4 and A8. The reproducibility of this superlubricity transition in A12 was further confirmed through additional friction measurements, as shown in Figure S2.

Importantly, control experiments were performed to rule out the possibility that the observed friction transition results from IL squeeze-out or depletion. We performed friction tests directly on the bare graphite substrate under identical loading conditions (Figure S3), which show a monotonically increasing friction–load relationship with a friction coefficient of ~ 0.198 , in stark contrast to the sharp friction drop observed in the $[\text{N}_{88812}][\text{A}_{12}\text{BMB}]$ system with a much lower friction coefficient of ~ 0.0013 in the high-load regime. This indicates that the IL film remains present and active at the graphite interface even under a high load. Moreover, AFM-IR results in Figure 2 would further confirm persistent ion layers at the IL–graphite interface at high loads, supporting that the friction transition stems from load-induced structural reorganization of the IL rather than the IL depletion.

Interfacial Chemical Change Examination

To probe whether the load-induced frictional transition observed for A12 stems from chemical changes at the sliding interface, we performed atomic-force-microscopy-based infrared spectroscopy (AFM-IR) under varying normal loads (cf. inset illustrations in Figure 2). This technique enables in situ detection of molecular vibrational fingerprints at the frictional interface while simultaneously applying a well-controlled normal load via the AFM probe. As shown in Figure 2, none of the three ILs (A4, A8, and A12) exhibited significant spectral shifts, new absorption bands, or intensity changes with increasing normal load. The characteristic peaks associated with $\text{C}=\text{O}$, $\text{O}=\text{C}=\text{O}$, $\text{C}=\text{C}$, $\text{C}-\text{H}$, and $\text{B}-\text{O}$ vibrations³⁹ remained unchanged in both position and shape, even at elevated loads corresponding to deep probe penetration into the ionic layers. These results indicate unambiguously that no chemical reaction, decomposition, or bond rearrangement occurs at the sliding interface, regardless of anion chain length or applied normal load. Importantly, the load-induced friction transition does not involve changes in the average alkyl-chain conformation but is instead associated with orientational rearrangements and increased configurational freedom within the confined interfacial layer, which do not produce distinct vibrational signatures detectable by AFM-IR. Therefore, the pronounced frictional drop observed for A12 under high load

(Figure 1) is not attributed to any tribochemical transformation. This conclusion is further supported by chemical state analysis via AR-XPS (Figure 3), which also reveals no chemical shifts or new species under varying probe depths.

During the friction measurements, a silicon nitride AFM probe slides across an IL film under different normal loads: low loads engage the outer, bulk-like layers, whereas high loads drive the probe deeper toward the interfacial region near the substrate.^{24,25} Therefore, to assess whether the frictional transition observed in A12 (Figure 1) is associated with any chemical change across different loads, we performed angle-resolved X-ray photoelectron spectroscopy (AR-XPS). This technique enables depth profiling by adjusting the photoelectron emission angle (θ): at smaller angles (e.g., 10°), the top bulk ions would be detected (Figure 3a), whereas larger angles (e.g., 80°) are more surface sensitive and sample near-surface ionic layers (Figure 3b). As shown in Figure 3(c)–3(e), the high-resolution spectra for all three ILs (A4, A8, and A12) exhibit no evidence of chemical shifts or new species formation across the full angular range (10° – 80°). The increase in peak full width at half-maximum (fwhm) with an increasing detection angle (10° – 80°) corresponds to a higher ion concentration near the surface at larger emission angles.⁴⁰ Specifically, the N 1s peaks appear consistently at ~ 402.1 eV for A4, A8, and A12, attributable to the N–C bond in the $[\text{N}_{88812}]^+$ cation. The B 1s peaks remain centered at ~ 188.1 eV across all samples, consistent with intact B–O coordination. These results corroborate the AFM-IR findings and confirm that the load-induced frictional behavior is not associated with changes in chemical composition or bonding.

The corresponding depth-dependent atomic concentrations of N (cation marker) and B (anion marker) are plotted in Figure 4(a)–4(c). A clear increase in both B and N species with increasing angle is observed for A4, A8, and A12 ILs at graphite surfaces, suggesting surface enrichment of the ions. While A4 displays pronounced vertical segregation with cations (N) enriched in the bulk at a smaller angle and anions (B) confined near-to-surface at a larger angle. In contrast, A8 exhibits surface-level charge balance ($\text{B} \approx \text{N}$) but with a cation-rich bulk, indicating intermediate segregation. Remarkably, A12 shows an approximately homogeneous cation–anion distribution across the entire depth range of the ILs on the graphite surface. This depth-level homogenization in A12 is likely a structural adaptation under high load, facilitating the observed drop in friction into the superlubricity regime. The vertical ionic redistribution observed for A12 strongly supports

the hypothesis of load-induced structural reconfiguration, rather than any chemical change, as the dominant contributor to its distinctive frictional behavior. The depth profiling results provide a valuable structural link that we further substantiated with nonequilibrium molecular dynamics (NEMD) simulations in the following section.

Mechanism

NEMD simulations were performed for the A12 IL on a single-layer graphene substrate, using a capped single-walled carbon nanotube as a simplified probe under varying normal loads, to elucidate the microscopic origin of the observed load-induced superlubricity transition. As shown in Figure 5, the simulated

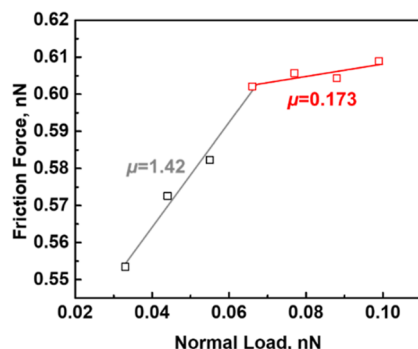


Figure 5. NEMD simulation of the friction of A12 under varying normal loads.

friction coefficient of A12 remains relatively high at low loads. Upon increasing the normal load, a pronounced drop in the friction coefficient is observed above a critical force threshold, consistent with the friction transition trend observed in the AFM experiment. The absolute values of friction force and normal load in the NEMD simulations differ from those measured in the AFM experiments. This disparity arises from the intrinsic differences in the system scale and boundary conditions: NEMD simulations are conducted on atomic scale systems with periodic boundaries and limited atom counts, whereas AFM experiments detect nano- to mesoscale contact regions with realistic probe geometries. Nonetheless, despite this difference in magnitude, the load-induced drop in friction

coefficient is fully consistent between the simulation and experiment.

We analyzed the orientational probability distributions of the two alkyl chains in the $[A_{12}BMB]^-$ anion (A12) with respect to the interface normal vector z under normal loads of 0.044 and 0.088 nN to elucidate the microscopic origin of the load-triggered drop in the friction coefficient. The orientation of each alkyl chain was quantified by calculating the angle between the chain vector and the interface normal vector (θ_1 , θ_2). As shown in Figure 6, at a lower normal load of 0.044 nN, both alkyl chains adopt a single, well-defined preferential orientation, indicative of a highly ordered interfacial arrangement. In contrast, at a higher normal load of 0.088 nN, each alkyl chain exhibits two distinct angular peaks, indicating multiple competing orientations and a transition to a less ordered, more flexible interfacial structure. This structural disorder at higher load may appear counterintuitive, but it is thermomechanically favorable. As discussed in prior studies,^{36,37,41} more ordered ionic layers promote stronger interfacial locking, which requires greater energy to shear or unlock. In contrast, the less ordered, partially reoriented structure reduces the shear resistance, facilitating the superlubricity observed in this work.

NEMD simulation in Figure S4 shows that all four alkyl chains in the $[N_{88812}]^+$ cation exhibit only minor shifts in their orientation angle distributions under two different loads. Unlike the anion, in which the orientational distribution splits into two distinct peaks under high load (Figure 6), the cation chains maintain a single-peak profile, indicating little change in the conformation. This confirms that the cation is mechanically stable and is unlikely to contribute significantly to the friction transition. Instead, the transition is primarily governed by the load-induced reorganization of the anion. The bulky $[N_{88812}]^+$ cation acts more like a structural backbone, while the flexible anion tails accommodate confinement and reconfigure to enable the emergence of superlubricity. At lower loads, the anion chains adopt a highly ordered orientation, forming a tightly interlocked interfacial structure that resists shear. As the load increases, these chains undergo conformational reorientation and become less ordered, resulting in a more flexible and dynamically disordered interfacial environment (cf. insets illustrated in Figure 6). This structural disorder reduces the

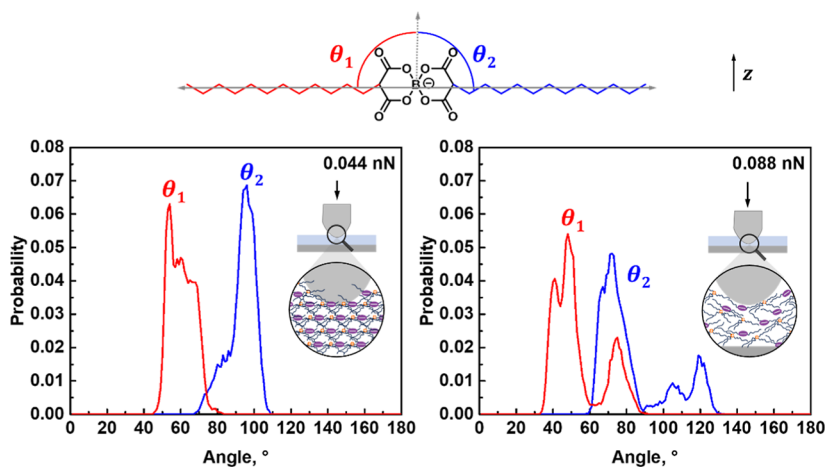


Figure 6. NEMD simulation: distribution of the alkyl chain orientation angles of the $[A_{12}BMB]^-$ anion in A12 with respect to the surface normal (z -axis) under low (0.044 nN) and high (0.088 nN) loads. A single sharp orientation dominates at low load, while a broader bimodal distribution emerges at high load. Insets illustrate the ions arrangements under varying loads.

energy required to disrupt the ionic layers during sliding,^{36–38} thereby facilitating the superlubricity. Such a load-responsive unlocking mechanism appears specific to the long-chain $[A_{12}BMB]^-$ anion, as the shorter-chain analogs lack the flexibility and conformational freedom necessary to support similar structural transitions.

CONCLUSIONS

We have demonstrated load-induced superlubricity without external stimuli. The long-chain borate IL $[N_{88812}][A_{12}BMB]$ undergoes a sharp transition from a higher friction at a low load ($\mu \approx 0.023$) to superlubricity ($\mu \approx 0.0013$) when the normal load exceeds ~ 30 nN (≈ 2.4 GPa), without electric fields, additives, or tribochemical film formation. This response is chain-length-dependent. The shorter-chain analogues, $[N_{88812}][A_4BMB]$ and $[N_{88812}][A_8BMB]$, exhibit stable ultra-low friction coefficients over the tested load range ($\mu \approx 0.0032$ and 0.0068), while only the longest-chain anion exhibits a load-triggered transition, highlighting the pivotal role of anion alkyl-tail length in enabling load-responsive interfacial structuring. Our data further identify this superlubricity transition as structural—rather than tribochemical—induced response. In situ AFM-IR and AR-XPS show no emergent vibrational features, chemical shifts, or new species under load, ruling out decomposition or tribofilm formation. Instead, depth-dependent elemental profiles indicate a near-homogeneous interfacial ion distribution for A12 at the interface. We propose a mechanical unlocking mechanism whereby confinement at high load reorganizes the interfacial ionic layers of A12 into a configuration with a low-shear plane formed by reoriented, interdigitated long alkyl chains, thereby collapsing friction into the superlubricity regime. NEMD simulations identify load-dependent reorientation of the long anion alkyl chains that reduces interfacial locking and shear resistance as the origin of the superlubricity.

These findings provide molecular-design guidelines for ILs targeting high-pressure boundary lubrication: long anion alkyl chains in borate frameworks promote load-responsive interfacial reorganization and superlubricity, offering a route to additive-free, field-free control of nanoscale friction. However, although demonstrated here at the nanoscale on graphite, future work should map the operating window (load, speed, temperature/humidity), assess substrate generality and long-term stability, and also explore scaling toward larger contacts. Combining this type of mechanical confinement with electrostatic tuning or surface chemistry may further expand the available superlubricity regime.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.5c10210>.

Details of synthesis procedure and NMR spectra of the three ILs employed in this work; 1H NMR spectra of the synthesized ILs; repeating friction–load measurements of IL–graphite interfaces; friction–load relationship measured by AFM at the bare graphite surface; and NEMD simulation of the distribution of the alkyl chain orientation angles of the $[N_{88812}]^+$ cation in A12 (PDF)

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Notes

The authors declare no competing financial interest.

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